

Convex Analysis and Monotone Operator Theory

Chapter 1: Background

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Sets and Basic Notation I

- A **set** is a collection of distinct objects
- Nonempty sets: $A, B, C, \dots$
- Power set: 2^A is the family of all subsets of A
- Set operations: union $A \cup B$, intersection $A \cap B$, complement A^c
- Cartesian product: $A \times B = \{(a, b) : a \in A, b \in B\}$

Key Definitions:

- 1 A **line segment** between $x, y \in \mathbb{X}$ consists of points of the form:

$$[x, y] = \{(1 - \alpha)x + \alpha y \mid 0 \leq \alpha \leq 1\}$$

- 2 Related notations:

$$]x, y] = \{(1 - \alpha)x + \alpha y \mid 0 < \alpha \leq 1\}$$

$$[x, y[= \{(1 - \alpha)x + \alpha y \mid 0 \leq \alpha < 1\}$$

$$]x, y[= \{(1 - \alpha)x + \alpha y \mid 0 < \alpha < 1\}$$

Single-Valued and Set-Valued Operators I

Definitions:

- ① **Single-valued operator:** $T : X \rightarrow Y$ maps every point $x \in X$ to exactly one point $Tx \in Y$
- ② **Set-valued operator:** $A : X \rightarrow 2^Y$ maps every point $x \in X$ to a set $Ax \subseteq Y$
- ③ An operator can be **single-valued but not everywhere defined**:

$$A : X \rightarrow 2^Y; \quad x \mapsto \begin{cases} \{Tx\}, & \text{if } x \in D; \\ \emptyset, & \text{otherwise} \end{cases}$$

where $D \subseteq X$

Single-Valued and Set-Valued Operators II

Graph and Inverse:

- The **graph** of A is:

$$\text{gra } A = \{(x, y) \in X \times Y : y \in Ax\}$$

- For $C \subseteq X$:

$$A(C) = \bigcup_{x \in C} Ax$$

- **Composition** $B \circ A$ (where $B : Y \rightarrow 2^Z$):

$$B \circ A : X \rightarrow 2^Z; \quad x \mapsto B(Ax) = \bigcup_{y \in Ax} By$$

Single-Valued and Set-Valued Operators III

Domain, Range, and Inverse:

- **Domain:** $\text{dom } A = \{x \in X : Ax \neq \emptyset\}$
- **Range:** $\text{ran } A = A(X)$
- **Inverse** A^{-1} : $\text{gra } A^{-1} = \{(y, x) \in Y \times X : (x, y) \in \text{gra } A\}$

Key properties:

$$\text{dom } A^{-1} = \text{ran } A$$

$$\text{ran } A^{-1} = \text{dom } A$$

Python Example: Operators

```
1 # Single-valued:  $T: R \rightarrow R, T(x) = 2x$ 
2 def single_valued(x):
3     return 2 * x
4
5 # Set-valued:  $A(x) = [-|x|, |x|]$ 
6 def set_valued(x):
7     return {"min": -abs(x), "max": abs(x)}
8
9 # Test
10 x = 3.0
11 print(f"T({x}) = {single_valued(x)}")
12 print(f"A({x}) = [{-abs(x)}, {abs(x)}] ")
```

Visual Guide: Types of Operators

Types of Operators

Single-Valued

$$T: X \rightarrow Y$$

Each x maps to exactly one $tx \in Y$

Set-Valued

$$A: X \rightarrow 2^Y$$

Each x maps to a set $Ax \subseteq Y$

Graph and Inverse

$$\text{Graph: } \text{gra } A = \{(x, y) \in X \times Y : y \in Ax\}$$

$$\text{Inverse: } A^{-1} = \{(y, x) \in Y \times X : (x, y) \in \text{gra } A\}$$

$$\text{Domain: } \text{dom } A = \{x \in X : Ax \neq \emptyset\}$$

Ordered Sets I

Binary Relations: Let A be nonempty and $\preceq$ a binary relation on $A \times A$.

Properties:

- ① **Reflexive:** $(\forall a \in A) \quad a \preceq a$
- ② **Transitive:** $(\forall a, b, c \in A) \quad a \preceq b \text{ and } b \preceq c \Rightarrow a \preceq c$
- ③ **Antisymmetric:** $(\forall a, b \in A) \quad a \preceq b \text{ and } b \preceq a \Rightarrow a = b$
- ④ **Connex:** $(\forall a, b \in A) \quad a \preceq b \text{ or } b \preceq a$

Ordered Sets II

Ordered Set Hierarchies:

Type	Requirements	Properties
Directed Set	(1), (2)	Partial order with universal bounds
Partially Ordered Set	(1), (2), (3)	Transitive, antisymmetric relation
Totally Ordered Set	(1), (2), (3), (4)	Every two elements comparable

Bounds: For $B \subseteq A$:

- **Upper bound:** $b \in B$ is an upper bound if $(\forall a \in B) a \preceq b$
- **Maximal element:** $a \in A$ is maximal if $(\forall c \in A) a \preceq c \Rightarrow c = a$

Zorn's Lemma (Fact 1.1):

Theorem

Let A be a partially ordered set such that every chain in A has an upper bound. Then A contains a maximal element.

This fundamental result is equivalent to the axiom of choice and has important applications in functional analysis and optimization.

Order Relations on a Set

Directed Set: (1), (2)

Partially Ordered Set: (1), (2), (3)

Totally Ordered Set: (1), (2), (3), (4)

1. Reflexive: $(\forall a \in A) a \leq a$
2. Transitive: $(\forall a, b, c) a \leq b \text{ and } b \leq c \Rightarrow a \leq c$
3. Antisymmetric: $(\forall a, b) a \leq b \text{ and } b \leq a \Rightarrow a = b$
4. Connex: $(\forall a, b) a \leq b \text{ or } b \leq a$

Nets and Generalized Sequences I

Motivation: Sequences indexed by $\mathbb{N}$ are insufficient for characterizing topological properties in general spaces. We use **nets** instead.

Definition: A **net** (or **generalized sequence**) in X indexed by A is an operator $(x_a)_{a \in A}$.

Key property: A is a **directed set**, meaning:

- Reflexive: $a \preceq a$ for all $a \in A$
- Transitive: if $a \preceq b$ and $b \preceq c$, then $a \preceq c$
- Directed: for any $a, b \in A$, there exists $c \in A$ with $a \preceq c$ and $b \preceq c$

Nets and Generalized Sequences II

Convergence of Nets: A net $(x_a)_{a \in A}$ converges to $x \in X$ if:

$$(\forall V \in \mathcal{V}(x))(\exists c \in A)(\forall a \in A) a \succeq c \Rightarrow x_a \in V$$

Subnet and Subsequence:

- A net $(y_b)_{b \in B}$ is a **subnet** of $(x_a)_{a \in A}$ via $k : B \rightarrow A$ if:

$$(\forall b \in B) \quad y_b = x_{k(b)}$$

with the property: $(\forall a \in A)(\exists d \in B)(\forall b \in B) b \succeq d \Rightarrow k(b) \succeq a$

- A **sequence** is a net indexed by $\mathbb{N}$

Nets and Generalized Sequences III

Nets vs. Sequences

Sequence: Indexed by $\mathbb{N}$

$$(x_n)_{n \in \mathbb{N}} = (x_0, x_1, x_2, \dots)$$

Limited: sequences insufficient for general topologies

Generalization

Net: Indexed by a Directed Set

$(x_a)_{a \in A}$ where A is a directed set

General: applies to all topological spaces

The Extended Real Line I

Definition: The **extended real line** is:

$$[-\infty, +\infty] = \mathbb{R} \cup \{-\infty\} \cup \{+\infty\}$$

with the order extended by:

$$(\forall \xi \in \mathbb{R}) \quad -\infty < \xi < +\infty$$

The Extended Real Line II

Arithmetic Rules (Remark 1.3):

- $+\infty + (+\infty) = +\infty$, $-\infty + (-\infty) = -\infty$
- $+\infty + \xi = +\infty$ for all $\xi > -\infty$
- $-\infty + \xi = -\infty$ for all $\xi < +\infty$
- Expressions like $+\infty - \infty$, $0 \cdot \infty$, ∞/∞ are **undefined**
- For $\alpha > 0$: $\alpha \cdot (+\infty) = +\infty$, $\alpha \cdot (-\infty) = -\infty$
- For $\alpha < 0$: $\alpha \cdot (+\infty) = -\infty$, $\alpha \cdot (-\infty) = +\infty$

The Extended Real Line III

Extended Real Line: $[-\infty, +\infty]$

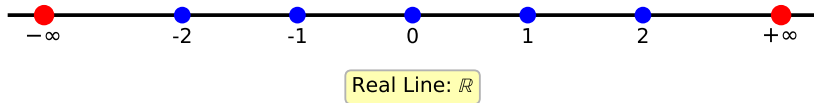


Figure: The extended real line: $[-\infty, +\infty]$

The Extended Real Line IV

Infimum and Supremum: For $S \subseteq [-\infty, +\infty]$:

- $\inf S$ is the greatest lower bound: $(\forall a \in S) \inf S \leq a$
- $\sup S$ is the least upper bound: $(\forall a \in S) a \leq \sup S$
- Special conventions: $\inf \emptyset = +\infty$, $\sup \emptyset = -\infty$

Functions with Values in the Extended Real Line I

Definition: Let $f : X \rightarrow [-\infty, +\infty]$.

The **domain** of f is:

$$\text{dom } f = \{x \in X : f(x) < +\infty\}$$

Key Sets Associated with f :

$$\text{gra } f = \{(x, \xi) \in X \times \mathbb{R} : f(x) = \xi\}$$

$$\text{epi } f = \{(x, \xi) \in X \times \mathbb{R} : f(x) \leq \xi\}$$

$$\text{lev}_\xi f = \{x \in X : f(x) \leq \xi\}$$

$$\text{lev}_{<\xi} f = \{x \in X : f(x) < \xi\}$$

Function Properties:

- f is **proper** if $-\infty \notin f(X)$ and $\text{dom } f \neq \emptyset$
- f is **increasing** (or monotone) if: $(\forall \xi, \eta \in D) \xi < \eta \Rightarrow f(\xi) \leq f(\eta)$
- f is **strictly increasing** if: $(\forall \xi, \eta \in D) \xi < \eta \Rightarrow f(\xi) < f(\eta)$

Functions with Values in the Extended Real Line III

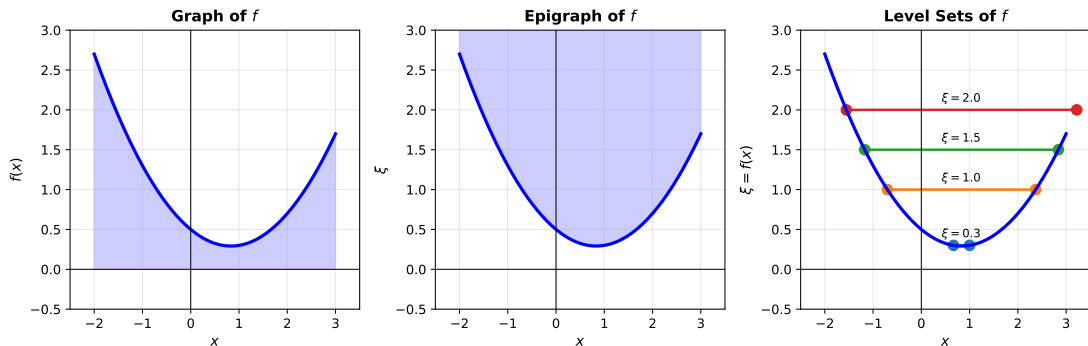


Figure: Graph, epigraph, and level sets of a function

Lemma 1.6: For a family of functions $\{f_i\}_{i \in I}$ from X to $[-\infty, +\infty]$:

$$\text{epi} \left(\sup_{i \in I} f_i \right) = \bigcap_{i \in I} \text{epi } f_i$$

$$\text{epi} \left(\min_{i \in I} f_i \right) = \bigcup_{i \in I} \text{epi } f_i \quad (I \text{ finite})$$

Topological Spaces and Basic Concepts I

Definition: A **topology** on X is a family $\mathcal{T}$ of subsets of X such that:

- 1 $\emptyset, X \in \mathcal{T}$
- 2 Arbitrary unions of sets in $\mathcal{T}$ are in $\mathcal{T}$
- 3 Finite intersections of sets in $\mathcal{T}$ are in $\mathcal{T}$

Key Definitions:

- **Open set:** an element of $\mathcal{T}$
- **Closed set:** complement of an open set
- **Neighborhood** of x : any open set containing x
- $\mathcal{V}(x)$: collection of all neighborhoods of x

Closure and Interior:

- **Closure** $\overline{C}$: smallest closed set containing C
- **Interior** C° : largest open set contained in C
- C is closed iff $C = \overline{C}$
- C is open iff $C = C^\circ$

Topological Spaces and Basic Concepts III

Closure vs. Interior

Interior: (a, b)
Closure: $[a, b]$

Open set: points at endpoints not included

Closed set: closure includes boundary points



Two-Point Compactification I

The Extended Real Line as a Compact Space: The extended real line $[-\infty, +\infty]$ equipped with the topology having as a base:

- Open real intervals (a, b)
- Intervals of the form $[-\infty, \xi)$ and $(\xi, +\infty]$

is a **compact Hausdorff space**.

Two-Point Compactification II

Fact 1.15: Let $(\xi_a)_{a \in A}$ be a net in $[-\infty, +\infty]$. Then:

- ① The nets $(\inf_{b \in A, a \preceq b} \xi_b)_{a \in A}$ and $(\sup_{b \in A, a \preceq b} \xi_b)_{a \in A}$ converge to $\underline{\lim} \xi_a$ and $\overline{\lim} \xi_a$, respectively.
- ② The net $(\xi_a)_{a \in A}$ possesses subnets converging to $\liminf \xi_a$ and $\limsup \xi_a$.
- ③ The net $(\xi_a)_{a \in A}$ converges iff $\liminf \xi_a = \limsup \xi_a$.

Two-Point Compactification III

Lemma 1.16: For nets $(\xi_a)_{a \in A}$ and $(\eta_a)_{a \in A}$ in $[-\infty, +\infty]$:

$$\underline{\lim} \xi_a + \underline{\lim} \eta_a \leq \underline{\lim} (\xi_a + \eta_a) \leq \overline{\lim} (\xi_a + \eta_a) \leq \overline{\lim} \xi_a + \overline{\lim} \eta_a$$

Lower Semicontinuity I

Definition 1.21: Let X be a Hausdorff space and $f : X \rightarrow [-\infty, +\infty]$.
 f is **lower semicontinuous** at $x \in X$ if:

$$(\forall \xi \in]-\infty, f(x)[)(\exists V \in \mathcal{V}(x)) \quad f(V) \subset [\xi, +\infty]$$

Equivalently, f is lower semicontinuous at x if:

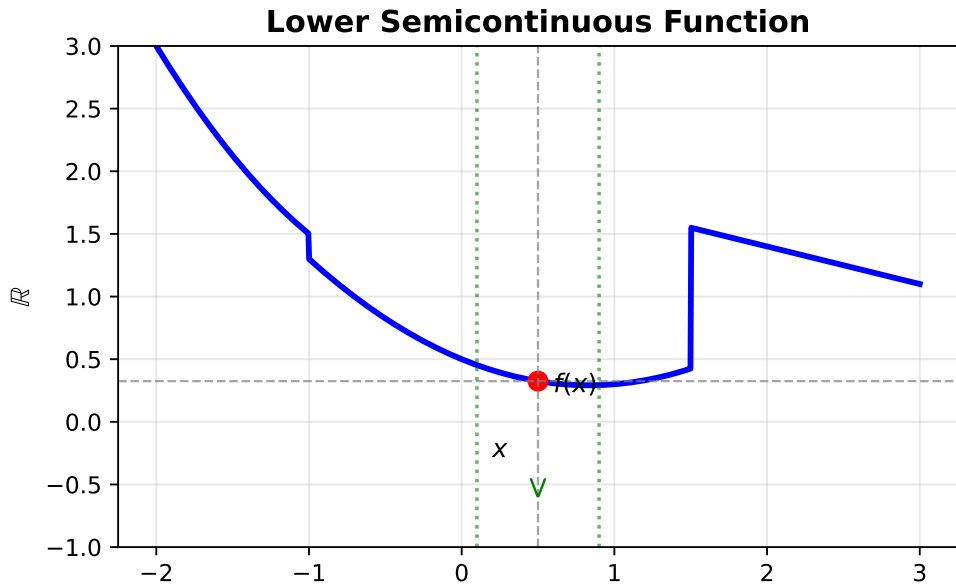
$$\liminf_{y \rightarrow x} f(y) \geq f(x)$$

Lower Semicontinuity II

Characterizations (Lemma 1.23):

The following are equivalent:

- ① f is lower semicontinuous on X
- ② $(\forall \xi \in \mathbb{R}) \quad \text{lev}_\xi f$ is closed
- ③ The epigraph $\text{epi } f$ is closed
- ④ For every $x \in X$: $f(x) = \sup_{(x_a)_{a \in A} \rightarrow x} \liminf f(x_a)$



Sequential Topological Notions I

Definition 1.33: A subset C of a Hausdorff space X is **sequentially compact** if every sequence in C has a sequential cluster point in C .

Note: Sequences are not sufficient to describe all topological properties in general spaces. For instance:

- A set can be closed but not sequentially closed
- Sequences may not adequately describe limit behavior in non-metrizable spaces
- Nets are the proper generalization

Sequential Topological Notions II

Sequential Continuity: A function $f : X \rightarrow Y$ is **sequentially continuous** at x if:

$$(\forall \text{ sequences } (x_n)_{n \in \mathbb{N}} \rightarrow x) \quad f(x_n) \rightarrow f(x)$$

Sequential Lower Semicontinuity: Replace nets with sequences in the characterization:

$$f(x) = \sup_{(x_n)_{n \in \mathbb{N}} \rightarrow x} \liminf_{n \rightarrow \infty} f(x_n)$$

Sequential Topological Notions III

Key Point: In metrizable spaces (e.g., $\mathbb{R}^n$), sequential notions coincide with topological ones. However, for general topological spaces, nets provide the proper framework.

Key Results Summary I

Major Theorems in Chapter 1:

- ① **Zorn's Lemma (Fact 1.1):** Every partially ordered set in which every chain has an upper bound contains a maximal element.
- ② **Compactness Characterization (Fact 1.11):** A set C in a Hausdorff space is compact iff every open cover has a finite subcover.
- ③ **Net Convergence (Fact 1.19):** If $(x_a)_{a \in A}$ converges to x and $(x_{k(b)})_{b \in B}$ is a subnet, then $x_{k(b)} \rightarrow x$.
- ④ **Lower Semicontinuity (Lemma 1.23):** Equivalence between continuity characterizations via neighborhoods and via epigraph closedness.
- ⑤ **Two-Point Compactification (Fact 1.15):** The extended real line $[-\infty, +\infty]$ is compact, and every net in it has convergent subnets.

Key Results Summary II

Computational Example: Function Analysis in Python

```
1 Define a lower semicontinuous function def f(x): return 0.3 * x**2 - 0.5 * x + 0.5
2 Check level sets and epigraph x = np.linspace(-2, 3, 100) y = f(x)
3 Compute infimum and supremum xi = 1.0 Height level level_set = x[y <= xi] x : f(x) <= xi
4 print(f"Level set at xi={xi}:
    [level_set.min() : .2f, level_set.max() : .2f]") print(f" Infimum of f : y.min() : .4f") print(f" Supremum of f : y.max() : .4f")
5 Verify lower semicontinuity numerically x0 = 0.5 eps = 0.01 neighborhood = x[np.abs(x - x0) < eps]
    f_neighborhood = f(neighborhood)
6 assert f(x0) <= np.min(f_neighborhood), "Lower semicontinuity violated!" print(" Function is lower semicontinuous at x0 = 0.5")
```

Key Results Summary III

Key Concepts Recap:

Concept	Definition
Domain of f	$\text{dom } f = \{x \in X : f(x) < +\infty\}$
Epigraph of f	$\text{epi } f = \{(x, \xi) : f(x) \leq \xi\}$
Level set	$\text{lev}_\xi f = \{x : f(x) \leq \xi\}$
Graph of operator	$\text{gra } A = \{(x, y) : y \in Ax\}$
Closure	$\bar{C}$ = smallest closed set containing C
Interior	C° = largest open set in C
Compact set	Closed and bounded in $\mathbb{R}^n$ (Heine-Borel)
Net convergence	$(x_a)_{a \in A} \rightarrow x$ if $(\forall V \in \mathcal{V}(x)) x_a \in V$ eventually

Preview: Chapter 2

Next Chapter: Hilbert Spaces

Chapter 2 builds on these foundational concepts to develop:

- Inner product spaces and Hilbert spaces
- Orthogonal complements and projections
- Separability and orthonormal bases
- Linear and nonlinear analysis in Hilbert spaces

The topological and operator-theoretic tools from Chapter 1 are essential for understanding:

- Convexity and subdifferentials
- Monotone operators
- Fixed-point theorems
- Optimization algorithms